

SUM Function in Excel

1. What is SUM?

The SUM function adds numbers together and gives you the total. It solves the everyday problem of totaling a list of values, such as daily sales, monthly expenses, or units in stock. A Data Analyst uses it to turn long columns of figures into totals that can be reported and compared.

2. Syntax

`=SUM(number1, [number2], ...)`

Argument	Required?	What it accepts
number1	Required	A number, a cell reference, or a range (for example B2:B6)
number2, ...	Optional	More numbers, cells, or ranges. Excel accepts up to 255 arguments.

The most common form is a single range, such as `=SUM(B2:B6)`. You can also add separate ranges, `=SUM(B2:B6, D2:D6)`, or mix a range with a number, `=SUM(B2:B6, 500)`.

3. Simple Example

Find the total sales for the week:

	A	B
1	Day	Sales
2	Monday	12000
3	Tuesday	15000
4	Wednesday	10000
5	Thursday	18000
6	Friday	14000
7	Total Sales	69000

Formula in B7: `=SUM(B2:B6)`

Excel reads the five values in B2 to B6 and adds them: 12000 + 15000 + 10000 + 18000 + 14000. **Result: 69000**

4. Another Example

Find the total units sold for each product across three months:

	A	B	C	D	E
1	Product	Jan	Feb	Mar	Total
2	Mouse	120	150	130	400
3	Keyboard	80	95	75	250
4	Headset	60	45	55	160

Formula in E2: `=SUM(B2:D2)`, then copied down to E4.

For Mouse, Excel adds the cells across the row: 120 + 150 + 130.

Result: 400. Copying the formula down gives 250 for Keyboard and 160 for Headset.

5. When to Use It

- Calculate total sales revenue for a day, month, or year
- Calculate total expenses or costs for a period
- Calculate the total number of orders or units sold
- Calculate total stock units in inventory
- Calculate row totals across months or quarters in a monthly report

6. Important Things to Remember

- SUM works with a single cell, a range, several ranges, or a mix of these. It adds down a column or across a row.
- Empty cells and text inside a range are ignored. They do not cause an error.
- Numbers stored as text (often left-aligned, sometimes with a green corner triangle) are ignored in a range, so the total comes out too low.
- If any referenced cell contains an error such as #N/A or #DIV/0!, the SUM result shows that error.
- SUM includes hidden rows and filtered-out rows in the total.
- SUM does not update itself if you add data outside the selected range. Check that the range covers all your data.

7. Common Mistakes

Mistake: Selecting an incomplete range, such as B2:B5 when the data goes to B6. **Fix:** Check the highlighted range after typing the formula and make sure it covers every data row.

Mistake: Including the total cell in its own range, such as `=SUM(B2:B7)` typed in B7. This causes a circular reference warning. **Fix:** End the range on the last data row, above the total.

Mistake: Getting a total that is too low because some numbers are stored as text. **Fix:** Convert them to real numbers (for example with Text to Columns) and check that they right-align.

Mistake: Adding cells one by one, such as `=B2+B3+B4+B5+B6`. It is slow and easy to get wrong. **Fix:** Use a range: `=SUM(B2:B6)`.

8. Data Analyst Tip

Use the AutoSum shortcut: select the cell below your numbers and press Alt + = (Windows) to insert a SUM formula instantly. To double-check a total, select the data range and look at the Sum shown in the status bar at the bottom of the Excel window. It should match your formula result.